

Employee Attrition Analysis — Executive Summary

Prepared for: HR Director
Dataset: IBM HR Analytics | 1,470 Employees
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The Business Problem

Every year, the company loses employees — and each departure carries a hidden cost: recruitment fees, months of lost productivity, and the knowledge that walks out the door. This analysis examined data from 1,470 employees to answer one question: **Can we identify, in advance, which employees are most likely to leave — and why?**

The short answer is **yes**.

What We Found

The company is losing approximately 1 in 6 employees (16.1% attrition rate). While this may appear manageable, the losses are not spread evenly — they are heavily concentrated in specific roles and driven by a small set of predictable, fixable factors.

Where the Problem Is Worst

Role / Department	Attrition Rate
Sales Representatives	~39% — nearly 2 in 5 leave
Lab Technicians	~24%
Human Resources roles	~23%
Sales Department (overall)	~20.6%
R&D Department	~13.8%

Sales Representatives represent the single most urgent retention crisis. At current rates, the company is effectively replacing its entire Sales Rep team every 2–3 years.

The Top 3 Reasons Employees Leave

After analysing every available data point — salary, age, department, job level, satisfaction scores, travel frequency, and more — three factors emerged as the strongest predictors of departure:

- Overtime.** Employees who regularly work overtime are significantly more likely to leave. Burnout, not pay, appears to be the primary driver of exits. This was the single most powerful signal in the entire dataset.
- Salary below the company median.** Employees earning less than their peers are at higher risk — particularly in roles where the external job market offers competitive alternatives. That said, salary alone does not tell the full story.
- Early tenure (first 0–2 years).** New joiners are the highest-risk group. Exit probability drops sharply after the 3-year mark, meaning the company is investing in onboarding and then losing people before that investment matures.

Does Salary Fix Everything?

No. Salary matters — employees who left earned roughly 28% less at the median than those who stayed. However, even among employees at similar pay levels, those working excessive hours and rating their work-life balance as "bad" showed far higher exit rates. Raising salaries without addressing workload and culture will not solve the problem.

Two Actions HR Can Take Immediately

- 1. Introduce an Overtime Monitoring Policy** Flag any employee who has worked overtime for three or more consecutive months for an automatic check-in with their manager. This costs nothing and directly addresses the #1 attrition driver. The goal is not to discipline — it is to intervene before a quiet resignation decision is made.
- 2. Launch a 24-Month Retention Programme for Sales Reps and Lab Technicians** Design a structured two-year onboarding and career development track for the two highest-risk roles. Include quarterly growth conversations, a visible promotion pathway with defined salary milestones, and mentorship pairing. Targeting early-tenure employees in these roles — before they leave — could realistically reduce exits in these categories by 15–20%.

An Important Caveat

This analysis is a powerful starting point, not a crystal ball. The model identifies *patterns* from historical data — it cannot see a difficult manager, a personal life change, or a competing job offer. **Use these findings to start better conversations, not to make decisions about individual employees.** A flag in a system should prompt a human connection — not a policy consequence.

Analysis conducted using IBM HR Analytics dataset. Three predictive models were trained and compared; the best-performing model correctly identified at-risk employees approximately 83–85% of the time when tested on unseen data.